



Store Performance Segmentation for Automotive Dealerships: A Data-Driven Approach to Optimize Advertising Efficiency

Identifying Distinct Store Segments Through Systematic Clustering Analysis

Author
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1. Introduction

Automotive dealerships with similar brand, market, and resources still show large performance differences. Traditional KPI reporting cannot reveal the latent factors driving this variation. We apply a systematic unsupervised learning framework to identify hidden dealership segments based on sales behavior, advertising efficiency, and operational characteristics. Using 2,557 stores over 24 months (61,322 store-month records), we evaluate 1,248 clustering pipelines across preprocessing, dimensionality reduction, and four clustering algorithms. Our results reveal a stable, interpretable three-cluster structure that explains major performance differences and supports more targeted resource allocation and marketing strategies.

2. Methodology

Data Preparation

[IsolationForest] → Remove 8% outliers (2,557 → 2,352 stores)

[QuantileTransformer] → Normalize to normal distribution

[PCA 95%] → Dimensionality reduction (11 → 5 components)

Algorithm Comparison

- K-Means (k=3)
- DBSCAN
- Hierarchical
- GMM

Evaluation Metrics

- Silhouette Score
- Davies-Bouldin Index
- Balance Ratio
- Stability (ARI, NMI)

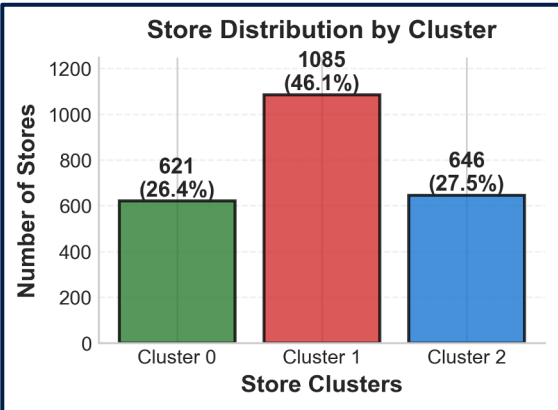
Final Selection: ISO_QT_PCA95_KMeans_k3

3. Key Results

Metric	Value	Target
Silhouette Score	0.586	≥ 0.5
Davies-Bouldin	0.706	≤ 1.0
Balance Ratio	1.75:1	≤ 3.0
Stability (ARI)	1.00	-

Three Clusters Identified

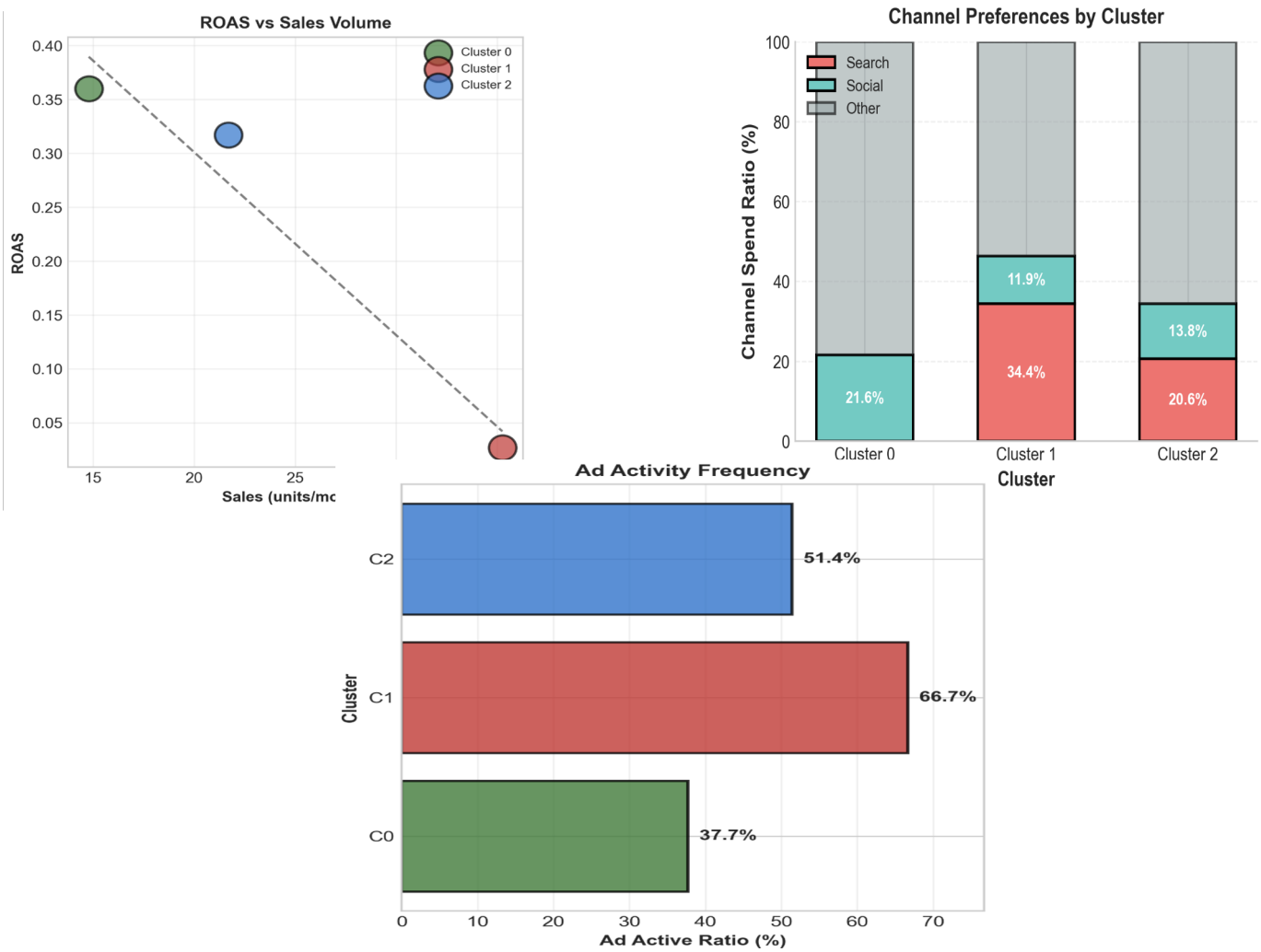
- **Cluster 0:** 621 stores (26.4%) - Low-Volume, High-ROAS
- **Cluster 1:** 1,085 stores (46.1%) - High-Volume, Low-ROAS
- **Cluster 2:** 646 stores (27.5%) - Medium-Volume, Balanced



4. Cluster Characteristics

Metric	Cluster 0	Cluster 1	Cluster 2
Sales (units/month)	14.79	35.25	21.70
Ad Spend (\$/month)	\$503	\$6,114	\$2,053
ROAS	0.360	0.027	0.317
Ad Active Ratio (%)	37.7%	66.7%	51.4%
Search Channel (%)	0.0%	34.4%	20.6%
Social Channel (%)	21.6%	11.9%	13.8%

5. Business Insights



6. Recommendations & Impact

The analysis identifies **Cluster 1** (1,085 stores, 46.1% of network) as the priority for optimization. These high-volume stores currently achieve the lowest ROAS efficiency (0.027) despite the highest ad spending (\$6,114/month). By reducing ad spending by 20-30%, optimizing search advertising strategies, diversifying channel mix, and improving conversion rates, we estimate monthly savings of \$1,200-\$1,800 per store, translating to \$1.33M - \$1.99M/month network-wide. The target ROAS improvement to 0.15-0.20 would represent a 5-7x increase in advertising efficiency. For **Cluster 0** (621 stores, 26.4%), the recommendation is to maintain current efficiency while scaling carefully, as these stores already achieve the highest ROAS (0.360). **Cluster 2** (646 stores, 27.5%) should pursue a balanced growth strategy, maintaining their current balanced performance across metrics while gradually expanding operations.

References:

- Rousseeuw, P. J. (1987). Silhouettes: A graphical aid to cluster analysis. Journal of Computational and Applied Mathematics, 20, 53-65.
- Lloyd, S. (1982). Least squares quantization in PCM. IEEE Transactions on Information Theory, 28(2), 129-137.
- Liu, F. T., et al. (2008). Isolation forest. Proceedings of the 8th IEEE International Conference on Data Mining, 413-422.
- Jolliffe, I. T. (2002). Principal Component Analysis. Springer-Verlag.



Overview

- Problem Statement & Research Objectives
 - Dataset Overview
 - Methodology: Systematic Clustering Pipeline
 - Three Store Clusters Identified
 - Key Business Insight - ROAS Efficiency Trade-off
 - Additional Business Insights
 - Discussion & Conclusion
 - Reference
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Problem Statement & Research Objectives

Problem Statement:

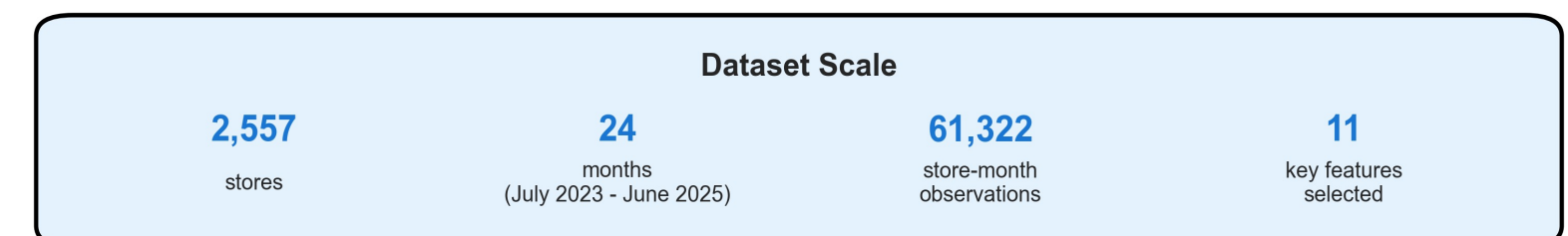
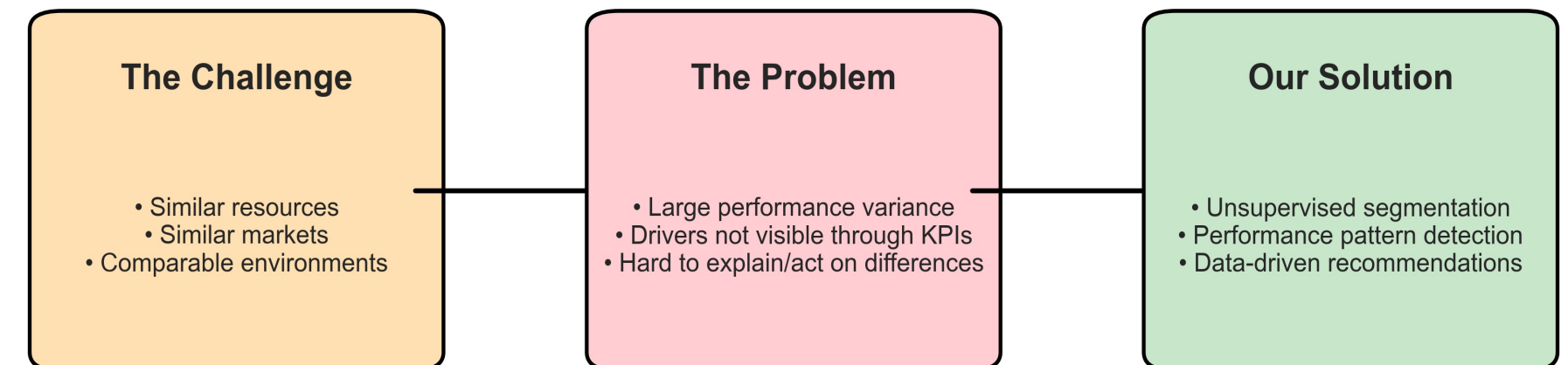
- Automotive dealerships have similar resources (same brands, similar markets, comparable resources)
- Yet show significant performance differences
- Traditional KPIs fail to explain these differences

Why This Matters:

- Cannot effectively allocate resources
- Cannot identify best practices
- Cannot optimize advertising strategies with precision

Research Questions:

1. How can we identify distinct store segments?
2. What characteristics differentiate performance?
3. How can insights guide business decisions?



Dataset Overview

Dataset Scale

- Stores: 2,557
- Time Period: 24 months
- Observations: 61,322 store-month records
- Features: 37 raw → 11 engineered

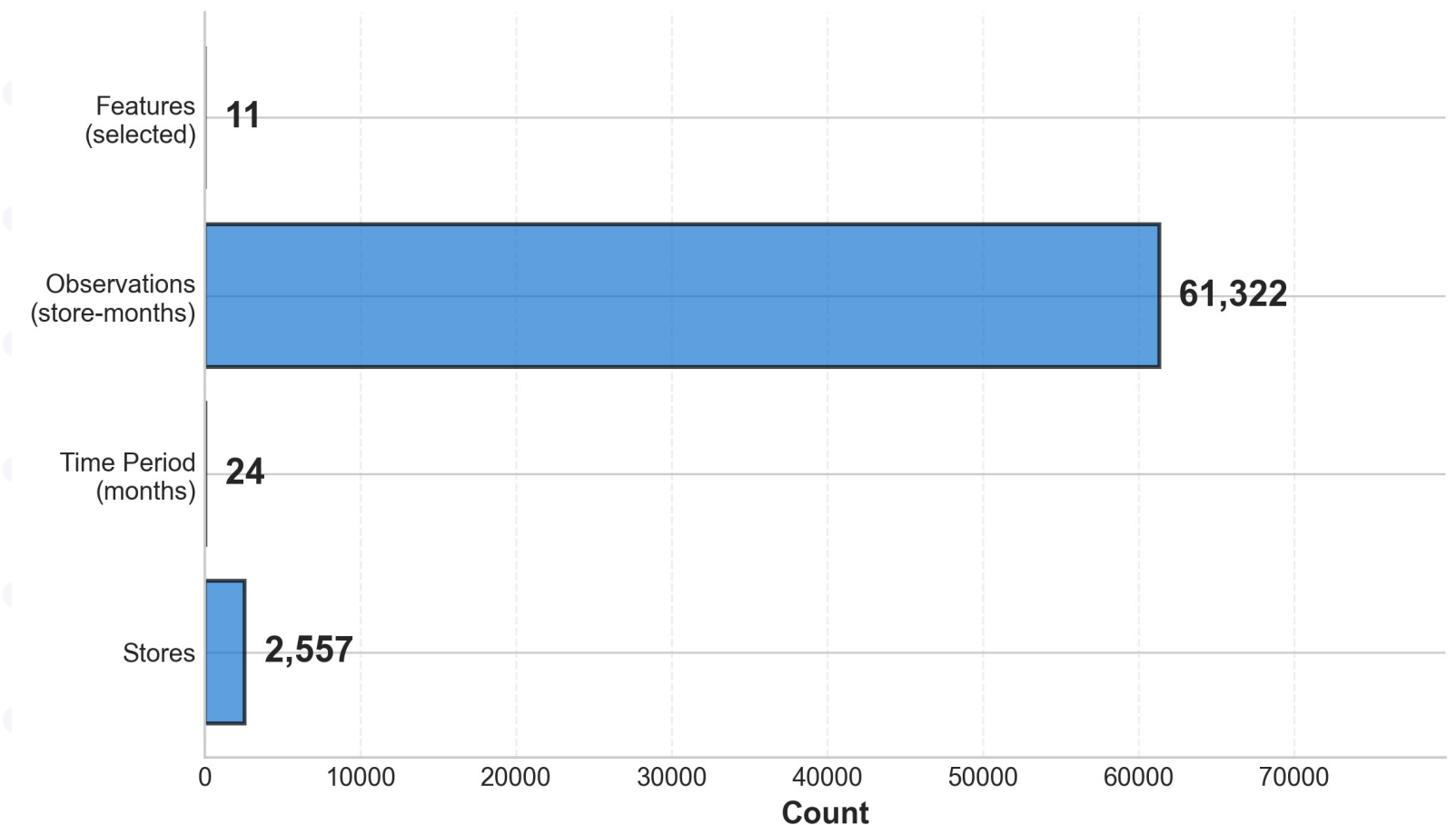
Data Quality

- Complete monthly time-series
- Multi-dimensional coverage:
 - sales, ad spend, web traffic, conversion metrics
- Cleaned + validated for missing values, anomalies

Feature Engineering

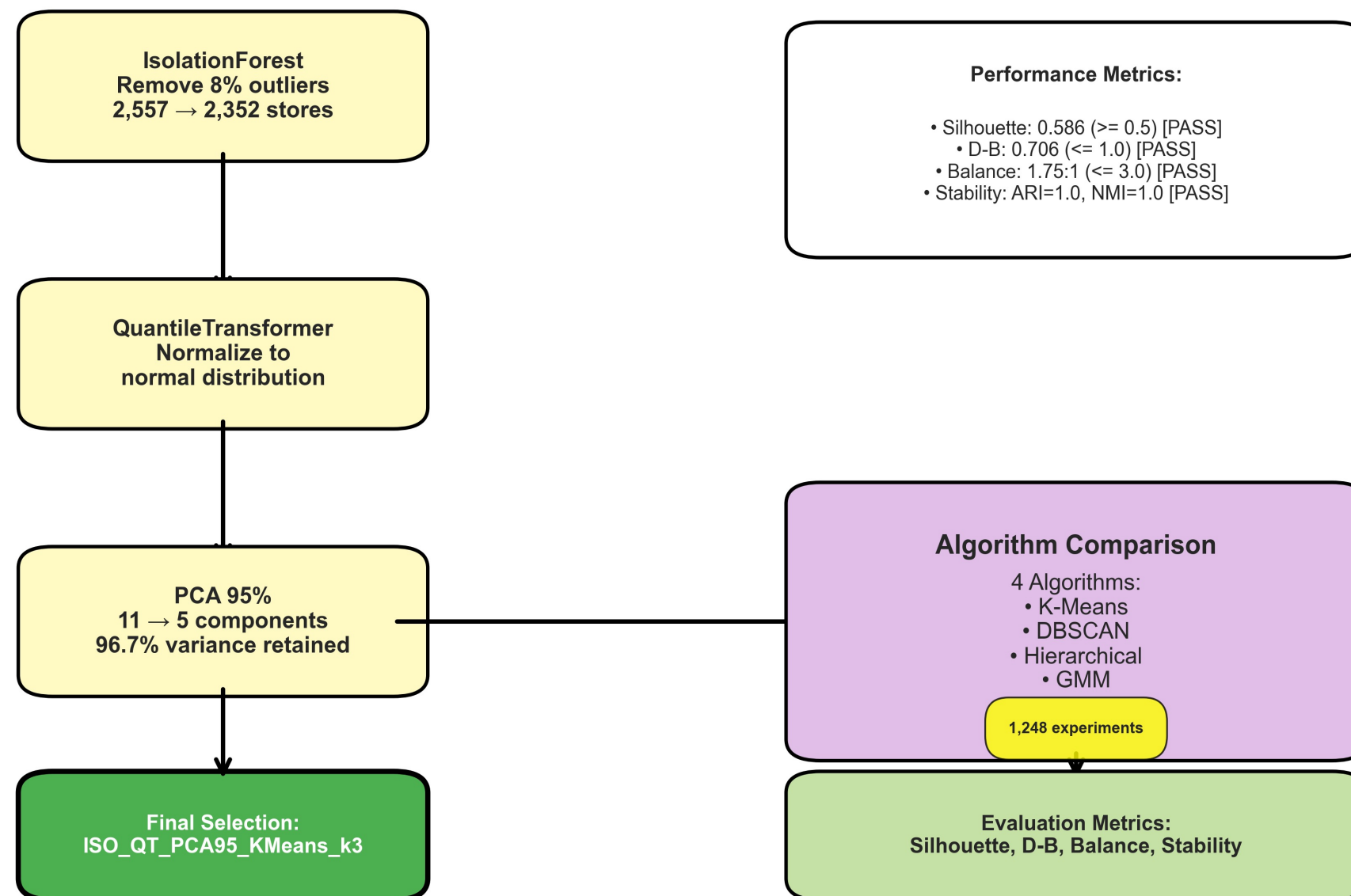
- 11 engineered features capturing ROAS, conversion rates, and advertising efficiency
- Lagged ad→sales features to model causal impact

Dataset Overview

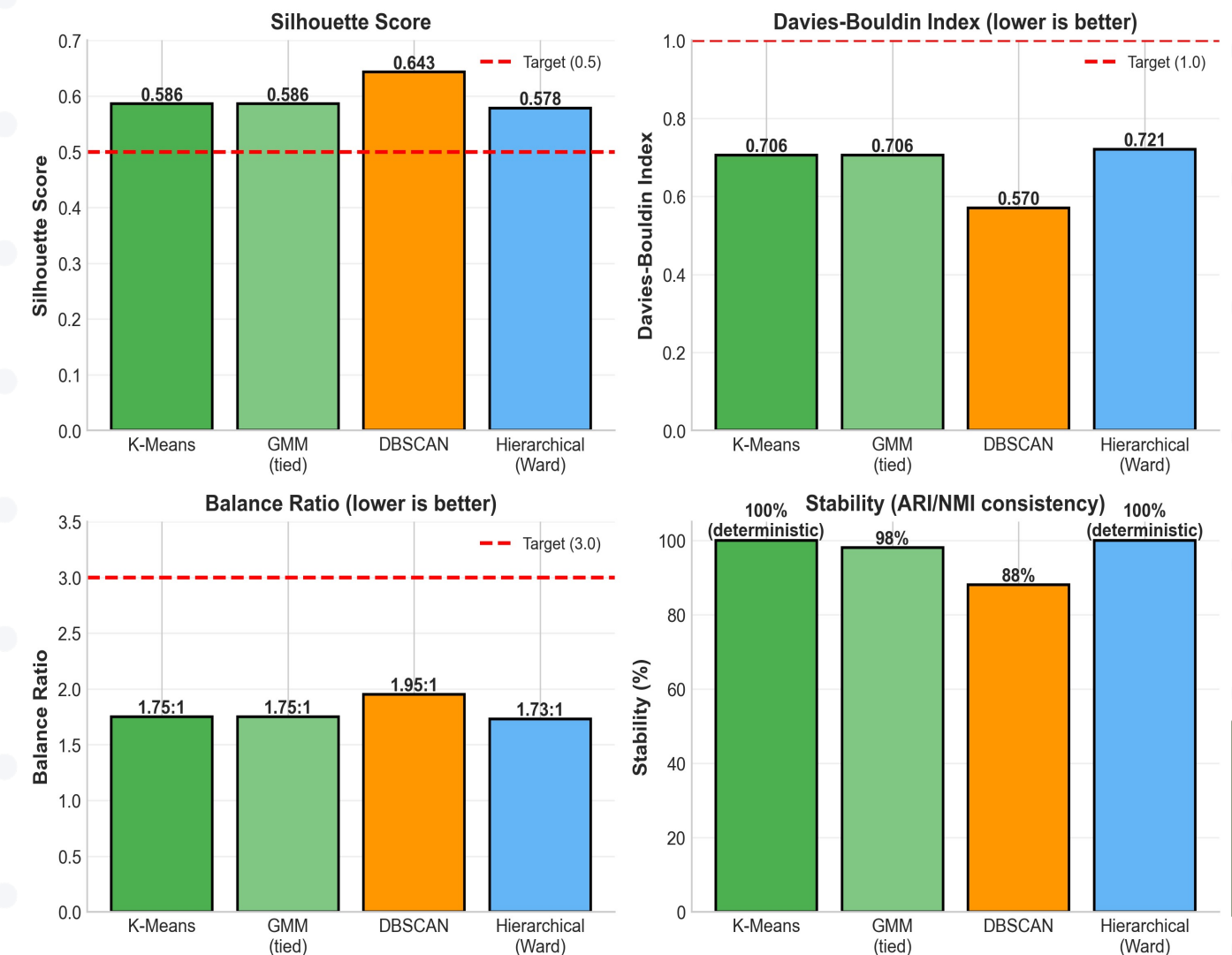


Methodology: Systematic Clustering Pipeline

From Raw Data to Actionable Insights



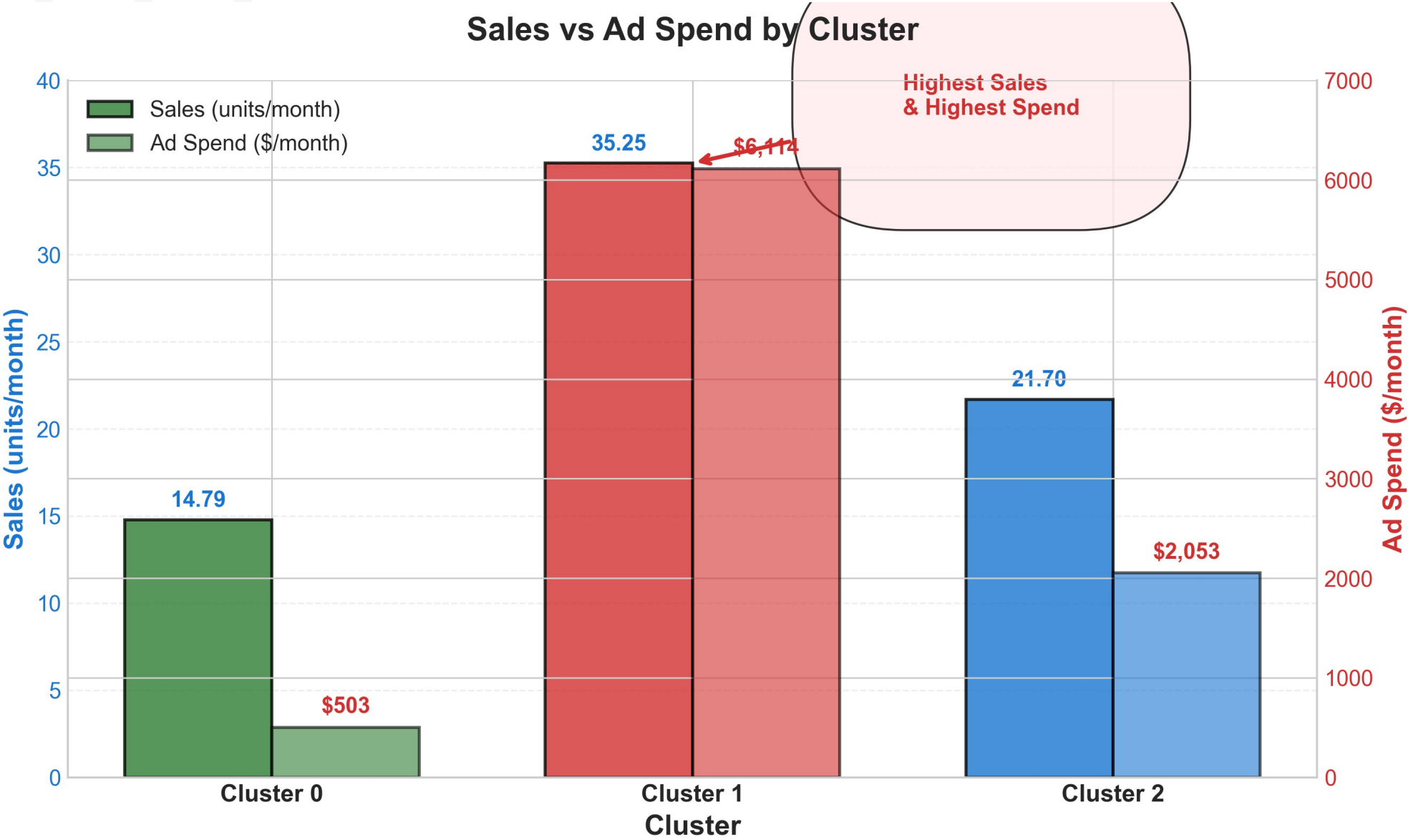
Algorithm Comparison: Performance Metrics



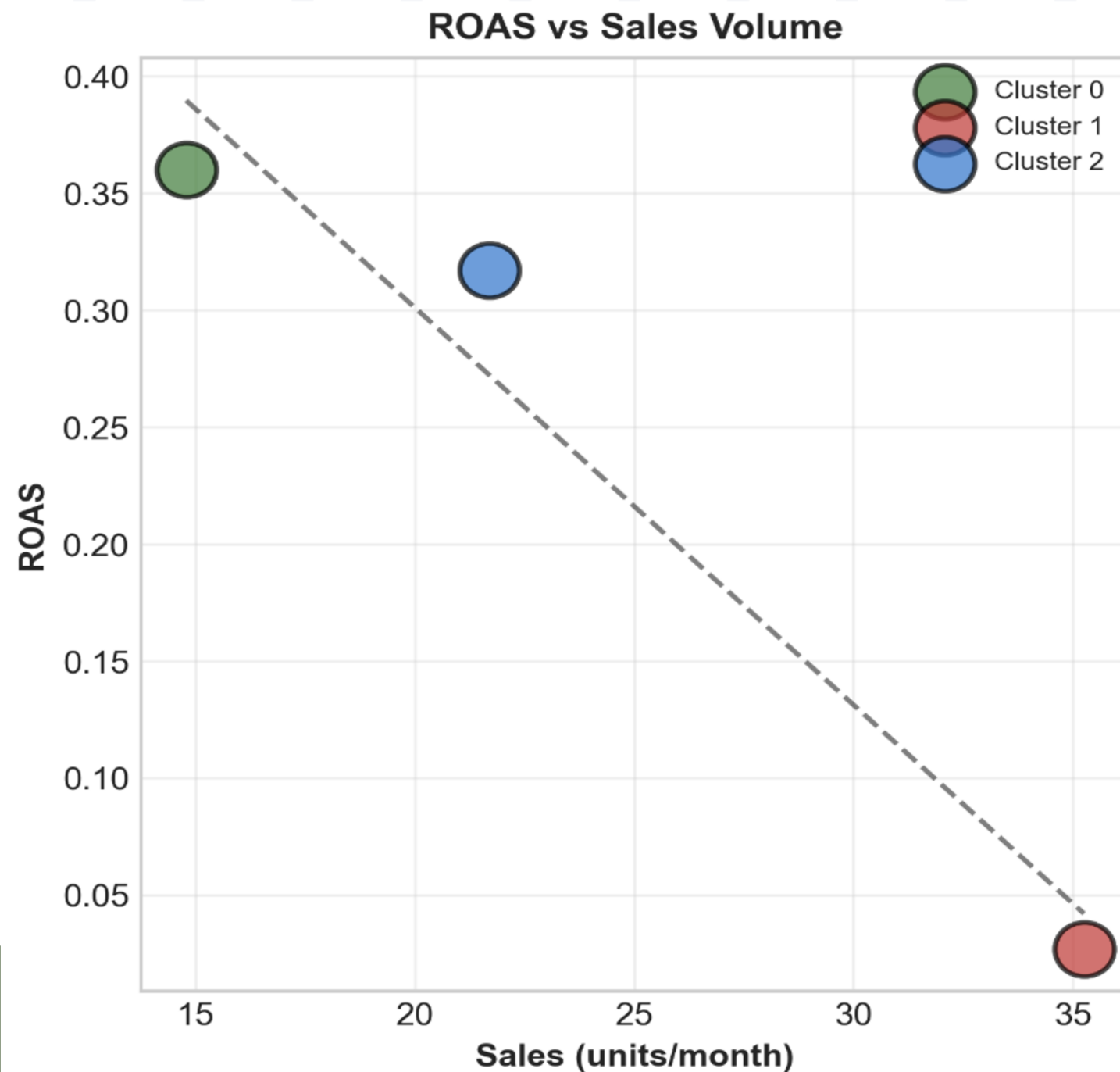
Three Store Clusters Identified

Cluster Comparison: Key Performance Metrics

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Key Business Insight - ROAS Efficiency Trade-off



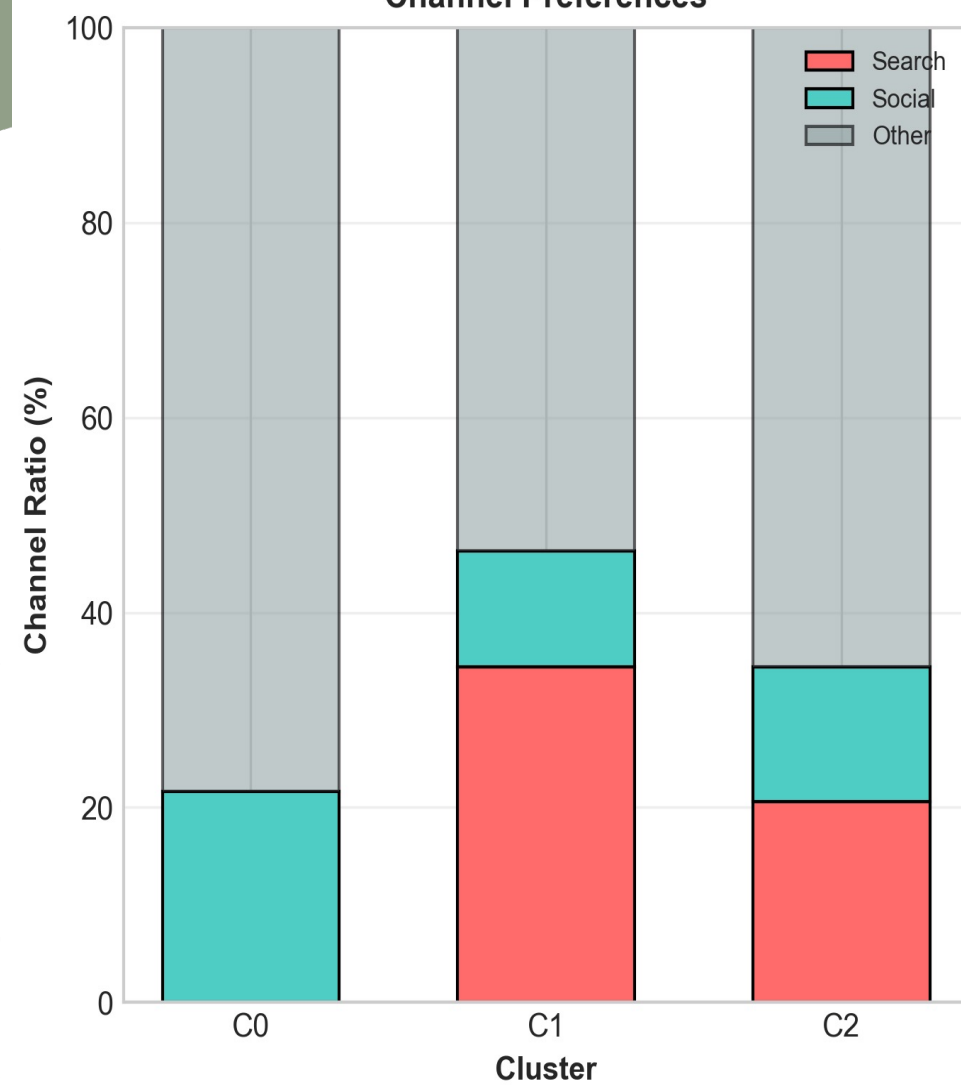
Inverse Relationship:

Higher sales volume $\neq$ Higher ROAS efficiency

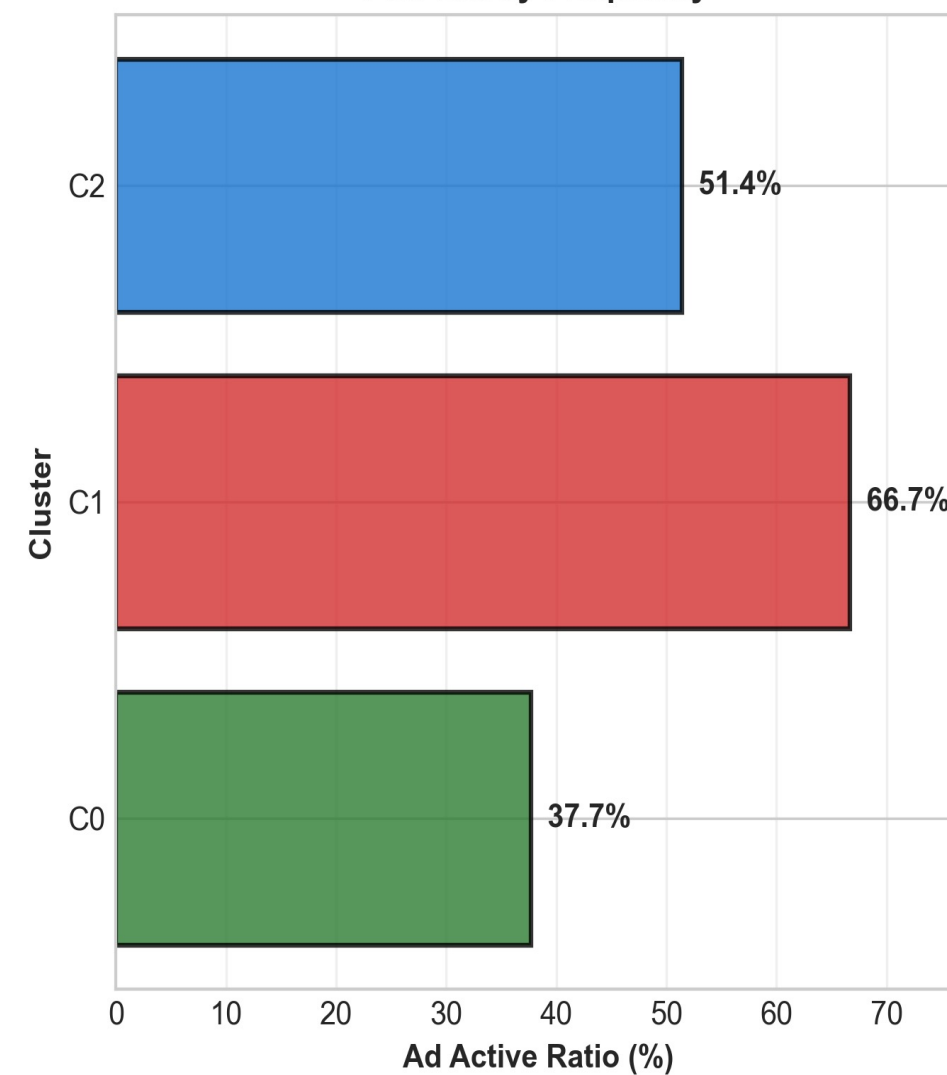
- Cluster 1: 35.25 units, ROAS = 0.027 (lowest)
- Cluster 0: 14.79 units, ROAS = 0.360 (highest)
- 13.3x efficiency difference

Additional Business Insights

Channel Preferences



Ad Activity Frequency



Channel Strategy Differences:

- Cluster 0: Social-heavy (22% social, 0% search)
- Cluster 1: Search-driven (34% search)
- Cluster 2: Balanced mix

Activity vs Efficiency:

- Cluster 1: 66.7% activity → ROAS 0.027
- Cluster 0: 37.7% activity → ROAS 0.360

Key Insight: More activity ≠ Better efficiency

Discussion & Conclusion

Key Findings

- Three distinct store segments emerged from systematic clustering
- High sales volume does not correspond to high advertising efficiency
- Cluster 1 drives the most sales but remains the least efficient (ROAS = 0.027)
- Cluster 0 achieves the best efficiency with much lower spend
- Store performance differences are driven by channel mix and ad behavior, not size

Interpretation

- Scale and efficiency are independent performance dimensions
- Traditional KPIs miss these deeper structural patterns
- A uniform strategy hides large optimization opportunities

Discussion & Conclusion

Implications

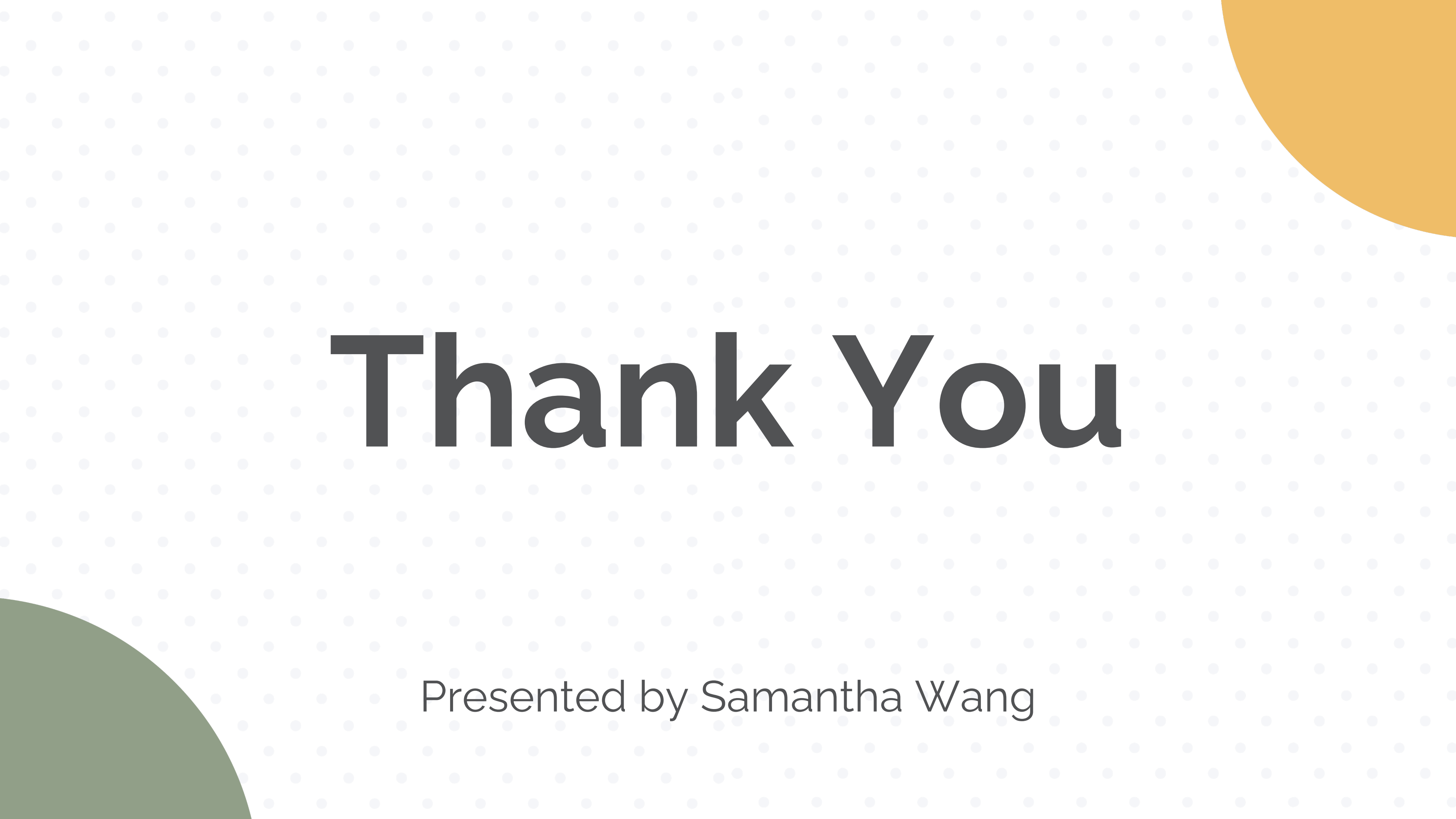
- Large optimization opportunity: \$1.33M–\$1.99M per month
- Nearly half the network (Cluster 1) can benefit from targeted ad adjustments
- Cluster-specific strategies outperform one-size-fits-all approaches

Limitations

- 24-month data window (long-term patterns not captured)
- Clustering is static (no time-evolving dynamics)
- External factors like market conditions not included
- Results validated on one industry only

Conclusion

- Our results show that data-driven segmentation uncovers structural performance patterns that traditional KPIs cannot capture. Using the ISO_QT_PCA95 pipeline, we identified stable and meaningful store segments that separate sales scale from advertising efficiency. Overall, the findings demonstrate that efficiency must be optimized independently—and that relying on a uniform strategy obscures substantial opportunities for improvement.



Thank You

Presented by Samantha Wang